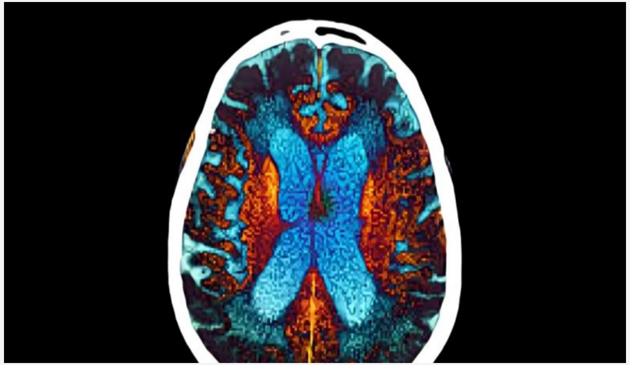
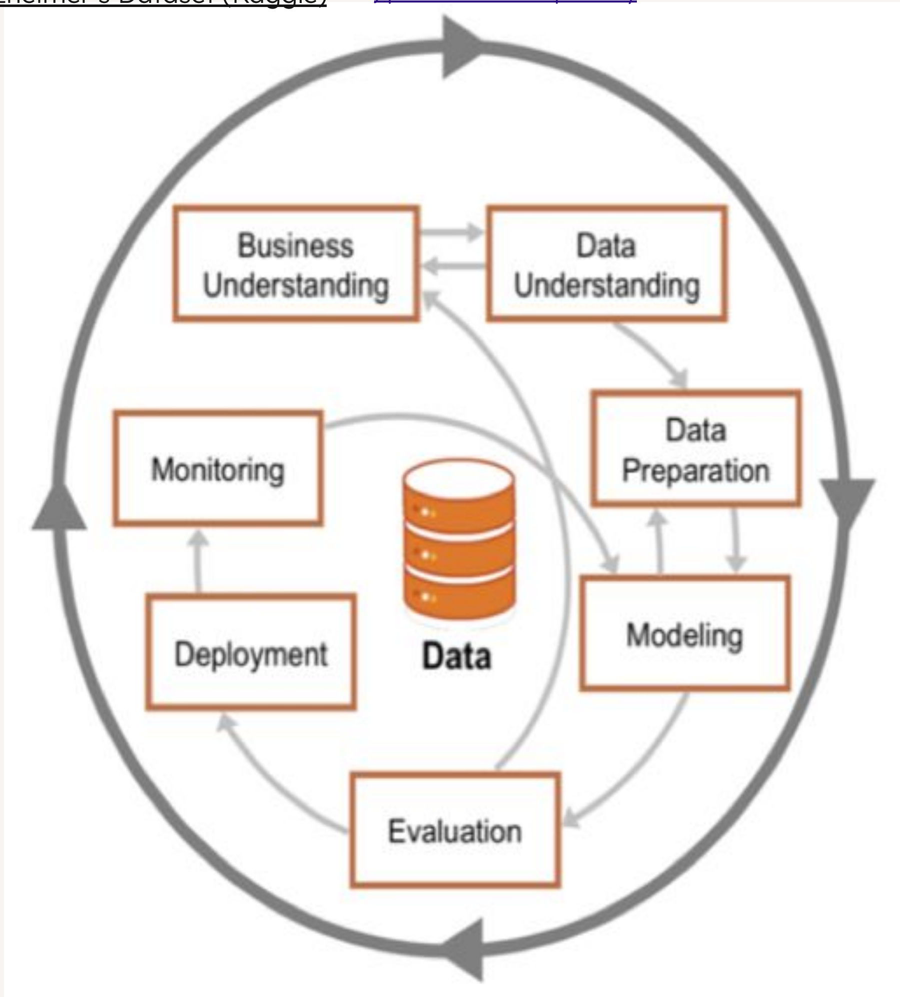
Predicting Alzheimer's Disease Risk from Clinical and Lifestyle Data



Jonathan (JT) Priest

Alzheimer's Dataset (Kaggle)   jtpriest-Github-Repository



Agenda

1. Business Understanding
2. Data Understanding & Data Preparation
3. Modeling, Evaluations, and Deployment
4. Key Findings and Next Steps

ML Objective, Use Case, and Stakeholders

ML Task

Binary Classification –

- Predict whether a person is labeled with an Alzheimer's diagnosis (Yes/No)
- Supervised binary classification using demographic, lifestyle, and clinical features
- Compare an interpretable baseline (Logistic Regression) to a higher-performing Random Forest model

Use Case

- Early-risk flag in a primary-care public-health setting
- Identify patients who should receive further cognitive testing
- Support resource planning for neurology / memory-clinic services

Primary Stakeholders

- Healthcare systems & clinics
- Public-health agencies / policy organizations
- Patients & caregivers – (indirectly, via earlier detection)

Business Understanding, Why Predict Alzheimer’s Risk?

- Alzheimer’s and dementia create large clinical and financial burdens for patients, families, and health systems
- Many patients are diagnosed late, when symptoms are severe and ongoing care is expensive
- A simple, data-driven risk model could help primary-care providers flag higher-risk patients for earlier cognitive assessment, not to replace a doctor but to support decisions

- Earlier risk identification creates a window for lifestyle changes and treatment to slow cognitive decline
- Helps identify high-risk patients earlier so clinicians can prioritize monitoring, lifestyle counseling, and cognitive screening

- Population-level risk scores help health systems plan clinic capacity and allocate neurology resources more efficiently
- Supports public health planning by quantifying how factors like age, genetics, and environment contribute to Alzheimer’s risk in different populations

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Data Understanding

Dataset Overview

Demographics	Lifestyle	Risk Factors	Cognitive	Country/ Region	Other/not tested for
Age Sex Education level	Smoking Alcohol Diet quality Physical Activity Sleep Social engages	Hypertension Diabetes Cholesterol BMI Depression Family History APOE- ε4 genotype	Test scores Cognitive testing Capacity	Childhood health Region effects Region and Country averages	Pre - existing health concerns Events causing major damage, such as brain or head injuries

Target Variable

Alzheimer's Diagnosis (Yes/No)

25 features, grouped (table)

Dataset Contains

74,283 individuals from 20 countries.

▶ Alzheimer’s Dataset (Kaggle) jtpriest-Github-Repository

Key Predictors from a Clinical Perspective

AGE

Strongest predictor; Alzheimer’s risk increases sharply with age

* Risk rises sharply after age 65.

APOE-ε4

genetic risk factor; “Yes” is associated with higher predicted risk

Family History

“Yes” also increases risk in the model

Cardio metabolic Factors

BMI, hypertension, diabetes, and cholesterol level act as vascular risk proxies

Lifestyle & Social

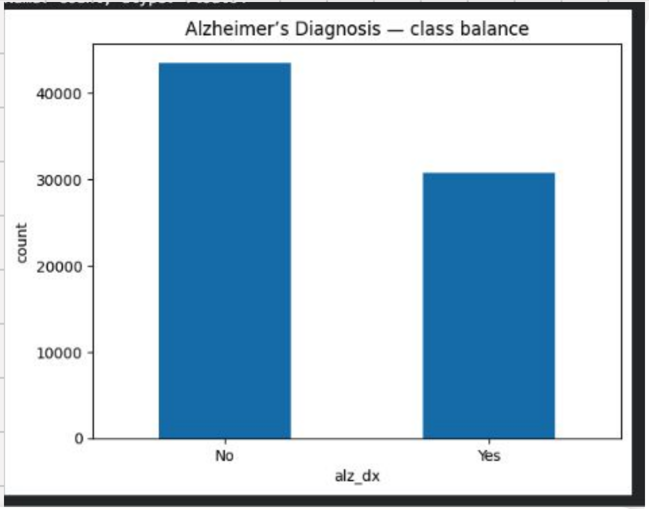
Physical activity, smoking, alcohol, diet, sleep, social engagement stress, and urban vs rural patterns can shift risk up or down

Cognitive Test Score

Summarizes current cognitive performance and is strongly related to diagnosis labels

The EDA and model largely confirm these expected risk factors

How Common is Alzheimer’s in the Data?

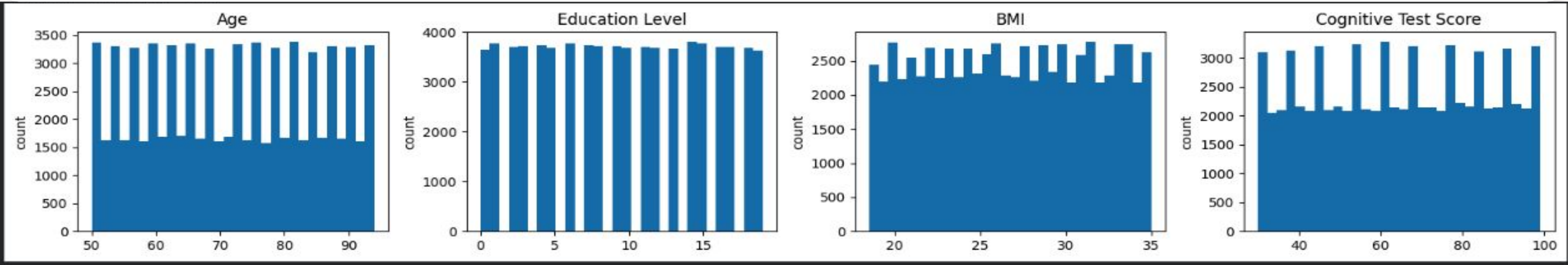


- **Total records: 74,283**
- **No diagnosis: 43,570 (~59%)**
- **Alzheimer’s diagnosis: 30,713 (~41%)**
- **Moderately imbalanced, but not extreme**

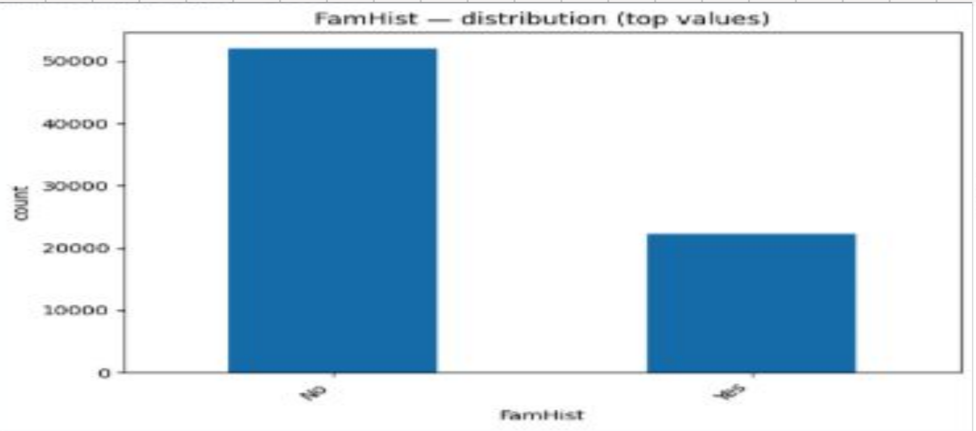
- I reported more than just accuracy – ROC-AUC, precision/recall, and confusion matrix to show trade-offs.
- ROC–AUC – how well the model ranks higher-risk vs lower-risk patients
- Precision and Recall – especially for the positive (Alzheimer’s) class
- Confusion matrix – to understand false negatives vs false positives

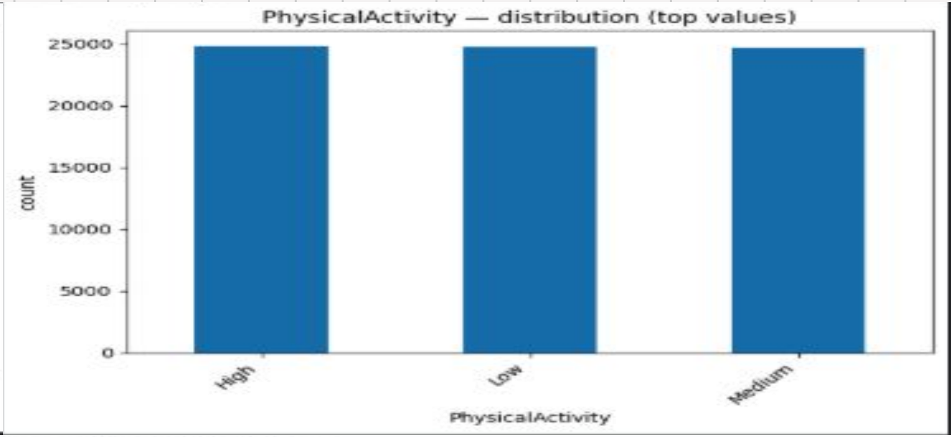
Exploratory Data Analysis (Numeric)

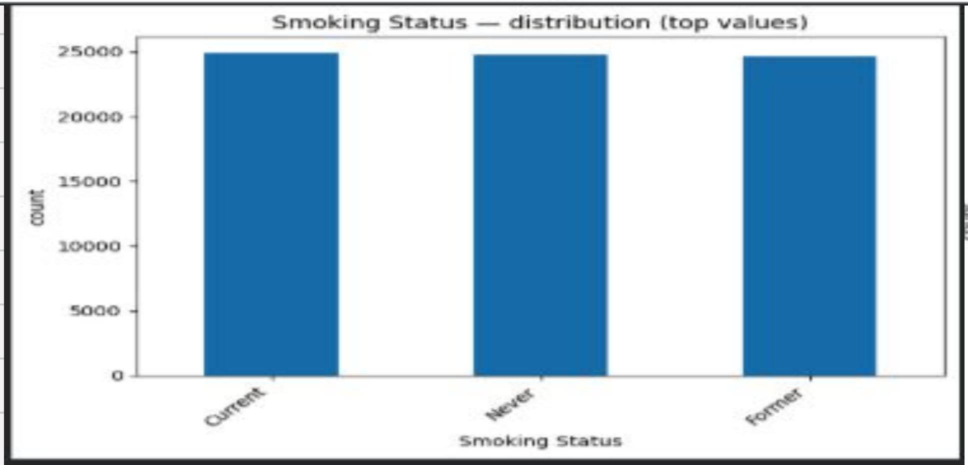
- **Age ranges roughly 50–95, with most individuals in their 60s–80s**
- **Education years are fairly spread, which can proxy for cognitive reserve**
- **Cognitive Test Scores range from 30–100, indicating a wide spectrum of baseline cognition**
- **APOE-ε4 and positive family history are less common but clearly show higher Alzheimer's rates in the crosstabs**

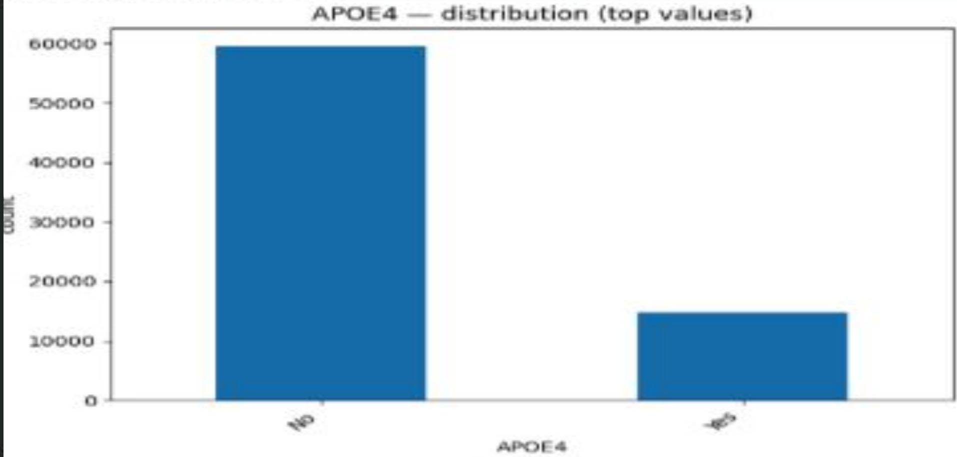


Exploratory Data Analysis (Categorical)









Training Pipeline

Logistic Regression training

- Logistic Regression has similar train and test performance -> it generalizes well

- Logistic Regression (overfitting check)
- Train ROC-AUC = .790

(Very similar to test, low overfitting)

Logistic Regression

- Train accuracy ≈ 0.716
- Train ROC-AUC ≈ 0.790
- Test ROC-AUC ≈ 0.794

Random Forest training

- Random Forest memorizes the training data (1.0 ROC-AUC) but still gives the best test ROC-AUC (~0.80)

- Random Forest (overfitting check)
- Train ROC-AUC = .801

(Bigger gap, model overfits training data)

Random Forest

- Train accuracy = 1.000
- Train ROC-AUC = 1.000
- Test ROC-AUC ≈ 0.801

- I chose Random Forest as the final model because it performs best on unseen data, while noting the overfitting risk

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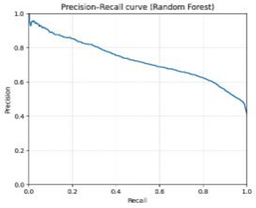
Modeling Pipeline



Baseline - Logistic Regression

- Interpretable linear model with regularization
- Test performance: Accuracy ≈ 0.72, ROC–AUC ≈ 0.79

Train/Test Split & Preprocessing

- 80/20 split, stratified by diagnosis to keep 41% positive rate in both sets
- Numeric features: StandardScaler
- Categorical features: OneHotEncoder with handle_unknown='ignore'
- All wrapped in a single scikit-learn Pipeline

Benchmark - Random Forest

- 300-tree Random Forest using the same preprocessed inputs.
- Test performance: Accuracy ≈ 0.73, ROC–AUC ≈ 0.80
- Selected as best model based on ROC–AUC

Alzheimer's Dataset (Kaggle) jtpriest-Github-Repository

Default vs Tuned Threshold

For a screening tool, it's safer to miss fewer true cases even if more people are flagged.

Default

Default .50 threshold

- Threshold = 0.50 (Random Forest)
- Accuracy ≈ 0.73
- Recall for Alzheimer's (positive class): ≈ 0.65
- About 35% of positive cases are missed (false negatives).

```
=== Metrics at threshold 0.500 ===
Accuracy: 0.7254
Confusion matrix:
 [[6809 1905]
 [2175 3968]]
```

Threshold

Tuned threshold (.377)

- Using the Precision–Recall curve, I tuned the threshold to maximize F1.
- Best threshold ≈ 0.377
- Recall jumps from ~0.65 → ~0.83, meaning it can catch more true Alzheimer's cases.
- Precision drops to ~0.61 and accuracy to ~0.71, meaning it can accept more false positives.

```
=== Metrics at threshold 0.377 ===
Accuracy: 0.71
Confusion matrix:
 [[5478 3236]
 [1073 5070]]
```

Feature Importance (Logistic Regression)

	feature	coef
0	num__Age	1.060019
62	cat__APOE4_Yes	0.639843
61	cat__APOE4_No	-0.630830
16	cat__Country_Russia	0.536592
11	cat__Country_India	0.479097
6	cat__Country_Brazil	0.454375
18	cat__Country_South Africa	0.431578
42	cat__FamHist_Yes	0.409350
41	cat__FamHist_No	-0.400337
13	cat__Country_Japan	-0.393576
21	cat__Country_Sweden	-0.392117
14	cat__Country_Mexico	0.383560
15	cat__Country_Norway	-0.346360
7	cat__Country_Canada	-0.344645
8	cat__Country_China	-0.133838
23	cat__Country_USA	-0.118988
4	cat__Country_Argentina	-0.099407
20	cat__Country_Spain	-0.088291
12	cat__Country_Italy	-0.084390
9	cat__Country_France	-0.073953

Interpretation

- I used the Logistic Regression coefficients for interpretability
- Age has the largest positive coefficient -> older age strongly increases predicted risk
- APOE-ε4 = Yes and positive family history both have strong positive effects
- APOE-ε4 = No and no family history have negative coefficients, lowering risk
- Some country effects likely capture unmodeled factors (healthcare systems, lifestyle) rather than causal differences

Risks & Mitigation

Major risks

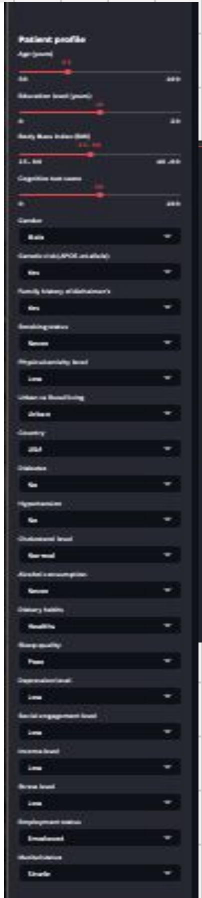
- False negatives: high-risk patients not flagged -> delayed diagnosis
- False positives: lower-risk patients flagged -> anxiety and unnecessary work-ups
- Dataset is synthetic and cross-sectional, not real EHR data
- Risk of bias across countries, income, or access-to-care features

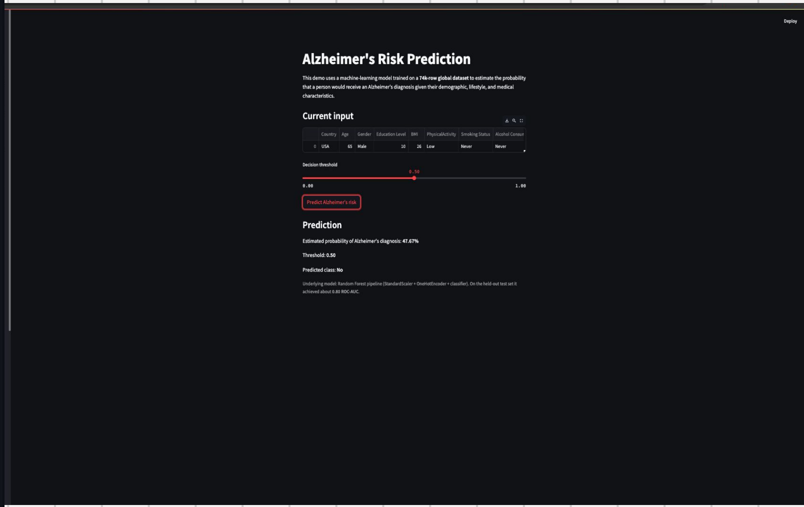
Mitigation Strategies

- Use the model as a decision-support tool, not as a standalone diagnostic
- Choose a recall-focused threshold for screening and monitor false-negative rates
- Before deployment, validate on real clinical data and recalibrate as needed
- Audit performance across subgroups and adjust or retrain if disparities appear

Deployment - Streamlit Risk Calculator

- Best model (Random Forest + preprocessing) saved as a scikit-learn Pipeline and loaded with joblib
- Built a Streamlit app that lets clinicians input a patient profile and view predicted risk in real time
- Threshold slider lets users explore the trade-off between sensitivity and false alarms.
- App clearly communicates that it’s a screening/risk estimation tool, not a medical diagnosis





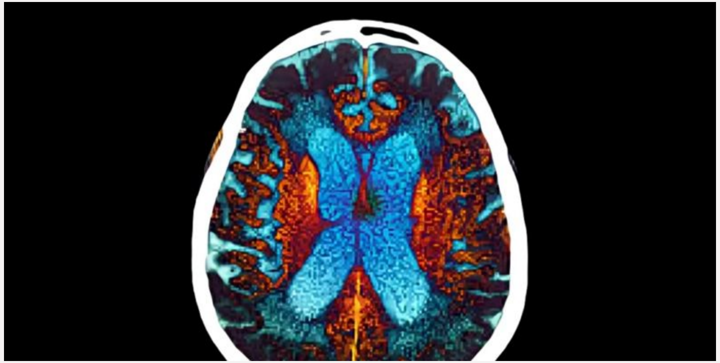
https://github.com/jtpriest1/Alzheimer-s-Prediction-Model

```
cd ~/Documents/alzheimers_demo
streamlit run alz_streamlit_app.py
```

Key Findings & Next Steps

- Using a 74k-record global dataset with 25 features, I can predict Alzheimer's diagnosis with ROC–AUC = 0.80 on a held-out test set.
- Random Forest slightly outperforms Logistic Regression, but both confirm expected risk factors: age, APOE-ε4, family history, and cardiometabolic / lifestyle variables
- Threshold tuning shows that I can raise recall from ~0.65 to ~0.83 by lowering the decision threshold, trading some accuracy for catching more cases
- I wrapped the model in a Streamlit app to demonstrate how a clinician could interact with the risk predictions
- Next steps: validate on real clinical data, check calibration and fairness, and explore SHAP-style explanations for individual predictions

Thank You!



Questions?